

Intelligent Food Delivery Analytics Using Live Swiggy Market Intelligence

23CSE452 Business Analytics — Individual Case Study

Akilan J · CB.SC.U4CSE23002 · CSE-A · Data collected 19 September 2026

1. Problem statement and objectives

Food delivery aggregators list thousands of restaurants whose ratings, prices, promised delivery times and promotional offers vary sharply from one locality to the next. Neither the platform nor its restaurant partners has a current view of which of these attributes actually move customer-facing quality or delivery performance, and because listings change daily, analyses built on static historical extracts describe a market that no longer exists.

Two decisions hang on this. Discounting is the most expensive lever a partner pulls and the most visible one a platform subsidises — if it does not travel with better ratings, both are funding something that does not buy loyalty. And if delivery time is driven by distance rather than by the restaurant, fixing it is a network-design problem for the platform, not a kitchen problem for the partner.

Objectives

- **Identify the drivers of restaurant rating** from publicly listed attributes, and establish how much of rating is explainable from listing data at all.
- **Model promised delivery time** and separate the contribution of last-mile distance from restaurant and city effects.
- **Benchmark competitors and segment the market** on price, rating, speed and discount depth, and convert the segments into pricing and operational recommendations.

2. Dataset source and collection method

Collection method: web scraping. Data was collected from www.swiggy.com/dapi/restaurants/list/v5, the public JSON endpoint that renders [swiggy.com/restaurants](https://www.swiggy.com/restaurants). It requires no login and exposes only business listing attributes already shown to any visitor. A purpose-built Python scraper (**scripts/scrape_swiggy.py**) issues one request per locality centroid for 70 localities — ten per city, spread across each urban area so the sample is not dominated by one high-income catchment — with a randomised 1.2–2.4 s delay between requests. Each response is flattened to one row per restaurant × locality probe.

Property	Value	Property	Value
Records collected	1,831 restaurant × locality rows	Unique restaurants	1,686
Cities / localities	7 / 70	Attributes	22 raw, 30 engineered
Records with a rating	99.8%	Authentication	None; no personal data

Important variables

Variable	Meaning	Variable	Meaning
avg_rating	Displayed rating, 1–5 (target)	delivery_time_min	Promised delivery time (target)
cost_for_two	Indicative price, from '₹400 for two'	last_mile_km	Distance from the probed locality
n_ratings	Rating count, from '5.6K+'	offer_type / offer_pct	Offer mechanic and depth
cuisines	Pipe-separated cuisine tags	is_chain	Brand has ≥ 3 outlets in sample

No ready-made Kaggle, UCI or GitHub dataset is used; the scraped data is the sole and primary dataset. The approved proposal had planned to add a historical Kaggle dataset and to collect via the Apify Swiggy Scraper. The Kaggle component was dropped because the course rules prohibit ready-made datasets as the primary source, and collection was done with an auditable in-repository scraper against the same public endpoint. Menu prices, Google ratings, weather and traffic were planned but are not obtainable — Swiggy's menu endpoint returns HTTP 202 to unauthenticated requests and the rest are not exposed publicly. **Ethics:** only business attributes are stored — no customer names, reviews or identifiers — and a crawl delay limits load on the source.

3. Data preparation and exploratory analysis

- **Display strings parsed to numbers:** '₹400 for two' → 400; '5.6K+' → 5,600.
- **Offers decomposed** from header and sub-header into four mechanics — capped percentage ('30% OFF UPTO ₹60'), flat rupee ('₹85 OFF ABOVE ₹199'), item price point ('ITEMS AT ₹69') and none. Parsing the header alone would collapse every price-point deal into an unusable 'ITEMS' label.
- **De-duplicated** 1,831 rows → 1,686 restaurants, after first deriving locality reach from the duplicates. **Outliers** winsorised (cost ₹120–₹900); delivery times outside 5–120 min dropped. **Missing:** 4 restaurants (0.2%) lack a rating and are excluded from rating models, leaving 1,667.



Figure 1 — Price and rating move in *opposite* directions (Spearman $\rho = -0.24$, $p < 0.001$).

Budget restaurants average **4.41**, mid-priced 4.28 and premium 4.31. Customers appear to rate against expectation: a ₹200 meal that arrives hot and correct satisfies, while a ₹700 meal carries a heavier burden of proof. Ratings are also compressed — the interquartile range spans about 0.3 of a star — so a modest R^2 on rating is expected and small coefficient effects remain commercially meaningful.

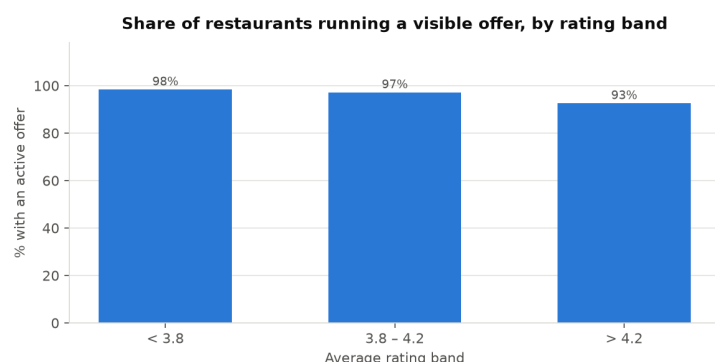


Figure 2 — Offer prevalence rises as rating falls ($\chi^2 = 15.8$, $df = 2$, $p = 0.0004$).

Discounting is near-universal — 94.1% of restaurants display an active offer — but it is not evenly spread. Restaurants rated below 3.8 are the most likely to be discounting and those above 4.2 the least. Median cost for two is ₹300 and median promised delivery 27 minutes; Pune is the fastest city at 23 minutes and Kolkata, Hyderabad and Bengaluru the slowest at 30.

4. Analytics method and implementation

Five methods from the Business Analytics syllabus were applied. All use a fixed seed (42), a 75/25 train-test split and 5-fold cross-validation.

Question	Method	Justification
What drives rating?	Multiple linear regression (OLS, HC3 robust SEs)	Coefficients are interpretable as business levers and support significance testing
Can a non-linear model do better?	Random Forest, Gradient Boosting + permutation importance	Captures interactions OLS misses; permutation importance ranks drivers honestly
Which restaurants are highly rated?	Logistic regression, unweighted and class-weighted	A classification framing management can act on, with threshold-free ROC-AUC
How do restaurants position?	K-Means, k by elbow and silhouette	Unsupervised segmentation with no pre-imposed labels
What drives delivery time?	OLS + Random Forest + Gradient Boosting	An interpretable minutes-per-km coefficient plus the best achievable predictor

4.1 Rating models

Rating was regressed on log cost, log rating count, delivery time, last-mile distance, cuisine count, locality reach, chain status, offer presence, offer depth and city dummies (n = 1,667). VIFs computed on the full design matrix peak at **2.67**, so multicollinearity is not a concern; Breusch-Pagan rejects homoskedasticity ($p = 1.8 \times 10^{-11}$) so HC3 robust standard errors are used. Significant effects: log cost -0.114 , offer presence -0.093 , delivery time $-0.008/\text{min}$, cuisine count -0.012 ; city effects are large (Kolkata $+0.132$, Pune -0.116). Rating count and chain status are not significant.

Rating model	CV R ²	Test R ²	RMSE	MAE
Linear Regression	0.251	0.261	0.203	0.163
Random Forest	0.469	0.487	0.169	0.131
Gradient Boosting	0.442	0.465	0.173	0.135

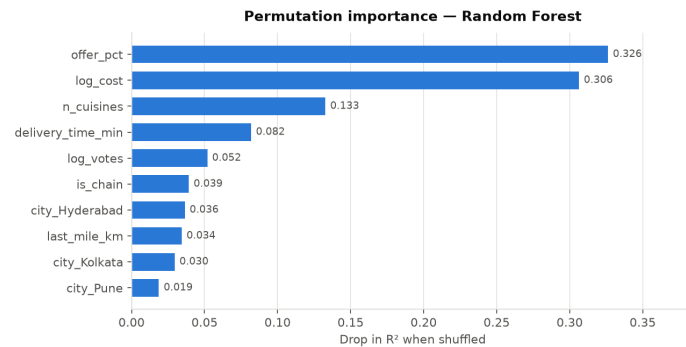


Figure 3 — Permutation importance, Random Forest rating model: offer depth outranks price.

Random Forest nearly doubles the linear model's explanatory power, confirming genuine non-linearity — but MAE 0.131 stars against an interquartile range of ~ 0.3 means listing attributes cannot substitute for knowing the food. A logistic model for rating ≥ 4.2 (79.5% of the sample) reaches ROC-AUC **0.726**; crucially, the unweighted version's 81% accuracy conceals just **14%** recall on under-performers, while class weighting raises that to **59%** at the same AUC — the usable model for a platform hunting weak listings.

4.2 Segmentation and competitor benchmarking

K-Means on standardised log cost, rating, delivery time, log rating count and offer depth. Silhouette is flat across k (0.205–0.234), indicating a continuum rather than sharply separated groups; k = 4 was retained for interpretability and is reported with that caveat (silhouette 0.215).

Segment	n	Rating	Cost for two	Delivery	Offer depth	% chain
Value performer	404	4.55	₹200	25 min	15.4%	89%
Premium performer	442	4.38	₹400	26 min	1.3%	80%
Premium laggard	267	4.22	₹400	35 min	4.6%	62%
Value laggard	554	4.19	₹300	28 min	62.9%	86%

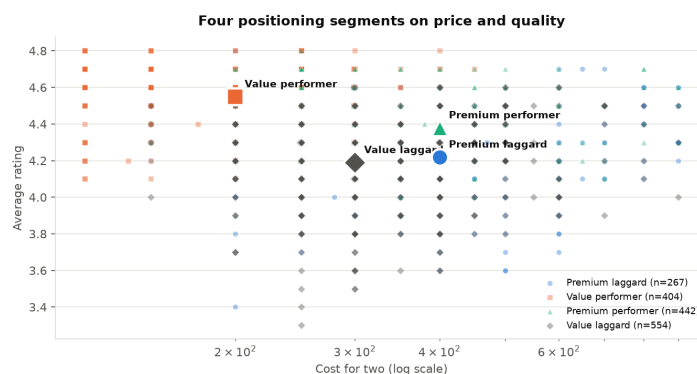


Figure 4 — The four segments on price and rating; shape and direct labels separate them.

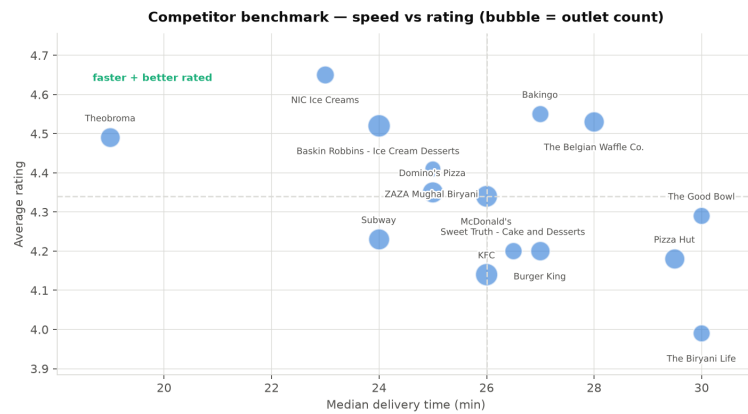


Figure 5 — Competitor benchmark of 15 multi-outlet brands; bubble size is outlet count.

Outlets were rolled up to the brand so multi-outlet operators compare as competitors. Dessert and beverage brands hold the favourable quadrant — NIC Ice Creams (4.65, 23 min), Theobroma (4.49, **19 min**), Baskin Robbins (4.52, 24 min) — while the large burger and pizza chains cluster lower on rating at comparable speed (KFC 4.14, Pizza Hut 4.18, Burger King 4.20) despite all running offers on essentially every outlet.

4.3 Delivery-time model

Delivery model	CV R ²	Test R ²	RMSE (min)	MAE (min)
Linear Regression	0.633	0.613	5.50	4.18
Random Forest	0.694	0.643	5.28	3.93
Gradient Boosting	0.692	0.651	5.22	3.94

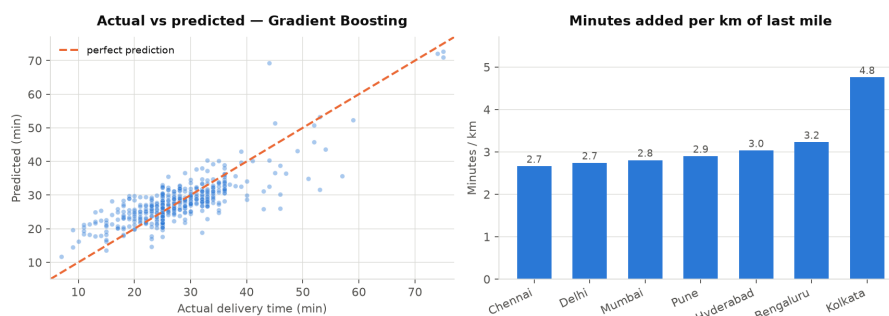


Figure 6 — Actual vs predicted delivery time, and minutes added per kilometre by city.

Delivery time is far more predictable than rating from the same fields. The interpretable OLS companion (R^2 0.640) gives the operational number: **each extra kilometre of last mile adds 3.43 minutes** ($p < 0.001$). Higher-rated restaurants are also faster (–5.72 min per rating point) and chains about 1.2 minutes quicker than independents. City effects are large and independent of distance — Pune is roughly 10 minutes faster than Kolkata at equal last-mile distance.

5. Comparison with three published studies

Three recent published studies on the same problem family are compared below. **Reported scores are not directly comparable across rows**: each study uses a different dataset, target variable, feature set and experimental setting, so a higher number does not by itself indicate a better method.

Published Study / Year	Dataset	Method Used	Evaluation Metric	Key Result	Comparison with Your Work
Garg, Ayaan, Parekh & Udandaraao (2025), arXiv:2503.15177	Indian-city food delivery dataset with traffic, weather, events and geospatial coordinates (secondary)	Linear Regression, Decision Tree, Bagging, Random Forest, XGBoost, LightGBM	R^2 , MSE	LightGBM best: $R^2 = 0.76$, MSE = 20.59	Closest comparator to \$4.3. Their higher R^2 rests on traffic, weather and event features that no public Swiggy endpoint exposes — a feature-availability gap, not evidence of a better method. We reach MAE 3.94 min from listing data alone and add an interpretable operational coefficient (3.43 min/km) they do not report. Strength: their richer feature set. Limitation: not reproducible from public data.

Published Study / Year	Dataset	Method Used	Evaluation Metric	Key Result	Comparison with Your Work
Turai, Praneetha, Aishwarya, Adil & Vangala (2025), WJARR 25(02), 1039-1046	Zomato restaurant and review data (secondary)	K-Means and Hierarchical clustering (k = 3) on cuisine and price; sentiment via Logistic Regression, Decision Tree, Naive Bayes, LDA	Cluster groups; classification accuracy	Logistic Regression and LightGBM best for sentiment; 3 cuisine-price segments	Same segmentation family as \$4.2, but they cluster on cuisine and price only and fix k = 3 with no stated selection criterion. We select k by elbow and silhouette and report it honestly (0.215, a continuum), and add delivery speed and offer depth — which is what surfaces the Value-laggard segment. Strength: they have review text for sentiment, which we lack. Limitation: no validation of cluster count.
Raj, Paul & Kumar (2025), IJRPR 6(5), 5962-5970	Zomato restaurant and user data (secondary)	SQL, Python, Power BI and Excel; EDA with logistic regression	Descriptive; no held-out predictive evaluation reported	Identified factors influencing ratings, cuisine popularity and location effects	Similar descriptive ambition to \$3, but the study stops at dashboards with no held-out validation, so its claims cannot be checked. We carry the same questions into tested models with train/test splits and cross-validation, and quantify effects ($\rho = -0.24$ price vs rating; $\chi^2 = 15.8$ offers vs rating band). Strength: broad BI tooling. Limitation: no predictive evaluation.

- **All three comparators use secondary datasets** — Zomato extracts or a compiled delivery dataset. This study is built on primary data collected first-hand, which is what makes the locality-level analysis possible.
- **None of the three examines discount depth against rating**, which is the finding reported in Section 6 and the contribution absent from the prior work reviewed.

6. Results, insights, and recommendations

Metric	Value	Metric	Value
Unique restaurants	1,686	Rating model — RF test R ²	0.487
Cities / localities	7 / 70	High-rating classifier ROC-AUC	0.726
Median cost for two	₹300	Delivery model — GB test R ²	0.651 (MAE 3.94 min)
Median delivery time	27 min	Minutes per extra km	3.43
Restaurants with an offer	94.1%	K-Means silhouette (k = 4)	0.215

Insight 1 — Deep discounting travels with lower ratings, not higher

Restaurants discounting 50% or more average **4.23** against **4.39** for everyone else — 0.16 stars on a scale whose interquartile range is about 0.3 (Welch $t = -13.3$, $p < 0.001$). Offer depth is also the highest-ranked feature in the Random Forest, and χ^2 confirms offer prevalence and rating band are not independent. **This is association, not proven causation**, and the likelier direction is that struggling restaurants discount to buy volume — but either way, discount depth reads as a distress signal, not a quality one.

Insight 2 — Price and rating are inversely related

Spearman $\rho = -0.24$; budget restaurants average 4.41 against 4.31 for premium, and controlling for everything else a one-log-unit rise in cost costs 0.114 rating points. Premium positioning carries a heavier burden of proof on a delivery platform than in a dine-in setting.

Insight 3 — Delivery is a network problem, not a kitchen problem

Distance dominates the delivery model at **3.43 minutes per kilometre**, and city effects are large and independent of it — Pune is about 10 minutes faster than Kolkata at equal distance. That points at fleet density and city operations, not restaurant preparation.

Insight 4 — Rating is only half-explainable from listing data

The best rating model reaches R² 0.487 while the delivery model reaches 0.651 on the same fields. Roughly half of rating variance lives in food quality, packaging and service consistency that no listing attribute captures.

Recommendations

- **Platform — treat discount depth as a risk indicator.** Route restaurants sustaining $\geq 50\%$ discounts into quality review rather than further promotional subsidy; the 554-strong Value-laggard segment is the target list.
- **Platform — deploy the class-weighted classifier.** At 59% minority recall it finds more than four times as many under-performing restaurants as the unweighted model at identical ROC-AUC.
- **Platform — attack delivery through network design.** At 3.43 min/km, shrinking the median last mile beats any restaurant-side intervention; prioritise Kolkata, Hyderabad and Bengaluru, which sit 4–7 minutes above Pune.
- **Platform — stop ranking on listing metadata alone** (R^2 0.487); feed order-level outcomes such as cancellations, complaints and reorder rate into any quality score.
- **Partners — do not buy rating with discounts.** Value performers reach 4.55 on ₹200 with 15% average depth; Value laggards spend 63% to reach 4.19.
- **Partners — fix speed before menu breadth.** Each minute of promised delivery costs 0.008 rating points and each extra cuisine tag 0.012 — breadth reads as lack of focus.

7. Conclusion and references

This study built a primary dataset of 1,686 restaurants across 70 localities and seven Indian metros by scraping Swiggy's public listing endpoint, then modelled rating and delivery time, benchmarked fifteen brands and segmented the market on price, quality and speed. The central outcome is that **the two questions a delivery business cares about are not equally answerable from public data**: delivery time is well explained (R^2 0.651, MAE under four minutes) and reduces to network design at 3.43 min/km, while rating is only half explained (R^2 0.487) because the half that matters lives in food and service quality that listings do not expose. The most commercially significant finding is that **promotional depth is negatively associated with rating** — 0.16 stars lower for restaurants discounting 50% or more — and that offer depth outranks price as a predictor. None of the three comparator studies examines this relationship.

Limitations and future work

A single-day cross-section cannot separate cause from effect; promised rather than actual delivery times are observed; traffic, weather and menu data are unavailable; the silhouette of 0.215 means the segments describe a continuum rather than natural kinds; and the endpoint returns roughly 28 restaurants per locality in Swiggy's own ranking order, so the sample skews toward restaurants the platform already promotes. Repeated collection over several weeks would turn this cross-section into a panel and allow the discount-rating question to be tested causally — observing whether ratings fall after discounts deepen, or the reverse.

References

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